

DEBI PRASAD PRADHAN

STAFF SOFTWARE ENGINEER | TECHNICAL ARCHITECT | ENTERPRISE PLATFORMS | CLOUD & GENAI ARCHITECTURE

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Staff Software Engineer and Technical Architect with 11+ years of experience architecting enterprise platforms, cloud-native systems, and GenAI-powered solutions. Proven track record of leading technical execution across multiple engineering teams, partnering with Principal Architects to transform business vision into scalable technical solutions, and delivering enterprise products from concept to production. Specialized in distributed systems, AWS architecture, performance engineering, cloud cost optimization, and AI-native platform development. Passionate about solving complex engineering challenges through modern architecture, technical leadership, and continuous product innovation.

Executive Highlights

- Led technical execution across 3 engineering teams (20+ engineers) while partnering with the Principal Architect to translate enterprise architecture into scalable technical solutions and successfully deliver enterprise platforms.
- Architected enterprise cloud-native platforms emphasising performance, scalability, reliability, security, and cloud cost optimisation.
- Reduced SDLC effort by 67% and accelerated engineering delivery by 3x through AI-powered Specification-Driven Development.
- Designed enterprise GenAI platforms leveraging OpenAI, Claude, Gemini, LangGraph, LangChain, Langfuse, MCP, and Knowledge Graphs.
- Built enterprise products from concept to production, consistently delivering customer-focused solutions within committed timelines.
- Led modernisation initiatives by reducing technical debt and aligning enterprise products with modern cloud-native and AI-native engineering practices.
- Designed AWS-based architectures supporting enterprise-scale scientific platforms, distributed systems, and large-volume file management.
- Established engineering standards, architecture governance, reusable frameworks, and technical best practices across multiple engineering teams.

Architecture & Engineering Expertise

- Enterprise Architecture
 - Cloud-Native Architecture | Enterprise Platform Architecture | API & Integration Architecture
 - Distributed Systems | Domain-Driven Design | Technical Modernisation
- Cloud & Platform Engineering
 - AWS Solution Architecture | Kubernetes & Docker | Performance Engineering
 - Cloud Cost Optimisation(FinOps) | Product Scalability | High Availability & Observability
- GenAI Engineering
 - Generative AI | Agentic AI | Model Context Protocol | LangGraph | LangChain | Langfuse
 - Gemini | Claude | OpenAI | SBERT | Prompt Engineering | Multi-Agent Systems
 - Retrieval-Augmented Generation | Semantic Search | Knowledge Graphs
 - Specification-Driven Development
- Leadership
 - Technical Leadership | Architecture Governance | Engineering Standards
 - Technical Roadmaps | Product Strategy | Customer Success | Technical Mentoring

PROFESSIONAL EXPERIENCE

Staff Software Engineer | Technical Architect

THERMO FISHER SCIENTIFIC | Bangalore, India

April 2022 - Present

Partner with the Principal Architect to define enterprise architecture, drive technical strategy, and lead technical execution across three engineering teams delivering enterprise platforms, cloud-native systems, GenAI solutions, and scientific data platforms. Responsible for architecture, engineering governance, cloud modernization, performance optimization, AWS solution design, engineering leadership, and successful delivery of enterprise-scale products.

Technical Leadership

- Led technical execution across 3 engineering teams (20+ engineers) while partnering with the Principal Architect to translate enterprise architecture into scalable technical solutions.
- Established engineering standards, architecture governance, reusable frameworks, and implementation best practices across engineering teams.
- Mentored software engineers and interns through architecture reviews, technical onboarding, and engineering best practices.
- Collaborated with Product Managers, Architects, UX teams, and business stakeholders to define technical roadmaps and platform strategy.
- Led customer-critical production issue resolution through architectural improvements and long-term platform stabilization.

Enterprise Architecture & Cloud Engineering

- Architected enterprise platforms from concept to production emphasizing scalability, resilience, maintainability, and long-term evolution.
- Designed AWS-native distributed systems using Kubernetes, Temporal, GraphDB, Aurora PostgreSQL, RabbitMQ, and modern event-driven architectures.
- Conducted enterprise-wide architecture assessments identifying platform standardization, reusable capabilities, and modernization opportunities.
- Designed an Enterprise Product Capability Graph providing visibility into products, services, APIs, ownership, dependencies, and technology relationships.
- Introduced Specification-Driven Development enabling AI-assisted engineering, reusable implementation specifications, and accelerated software delivery.

Performance, Cost Optimization & Delivery

- Designed AWS architectures optimized for scalability, performance, resilience, and cloud cost optimization.
- Optimized platform performance through distributed system design, asynchronous processing, scalable data access, and cloud-native architecture.
- Designed enterprise file management supporting secure upload and lifecycle management of files up to 1 TB.
- Reduced technical debt by modernizing legacy products and aligning engineering practices with modern cloud-native and GenAI architectures.
- Delivered enterprise products from concept to production while consistently meeting business commitments and maintaining engineering quality.

EDUCATION

- Master of Computer Applications (MCA) — Utkal University
- Bachelor of Computer Applications (BCA) — Utkal University

SELECTED ENTERPRISE INITIATIVES

Workflow Forge - AI-Powered Specification-Driven Development Orchestrator Platform

- Architected an AI-powered engineering platform transforming business specifications into AI-assisted implementation, testing, validation, documentation, and deployment workflows.
- Designed distributed workflow orchestration using Temporal integrated with Jira, GitHub.
- Reduced SDLC effort by 67% while accelerating engineering delivery by 3× through AI-powered Specification-Driven Development.

Analytical Data Service - Enterprise Scientific Knowledge Graph Platform

- Architected a scientific data platform managing proteins, molecules, analytical chemistry, and experimental datasets.
- Designed ontology-driven Knowledge Graph models and scalable ingestion pipelines using GraphDB, Apache NiFi, RabbitMQ and Python ETL for enterprise scientific data management.
- Designed semantic similarity search using SBERT embeddings and OpenAI models to improve scientific data discovery and relationship analysis.

PyWizard - Enterprise GenAI Scientific Intelligence Platform

- Architected a multi-agent GenAI platform automating scientific article interpretation, structured evidence extraction, and questionnaire generation.
- Designed LangGraph orchestration using OpenAI, Claude, Gemini, and Langfuse.
- Implemented provenance-aware AI validation and human-in-the-loop review workflows.
- Reduced scientific model preparation from weeks to hours through AI-assisted document understanding and automated questionnaire generation.

TECHNICAL SKILLS

Programming Languages - Java | Python | SQL

Enterprise Architecture - Enterprise Platform Architecture | Solution Architecture | Product Architecture | Cloud-Native Architecture | Distributed Systems | Event-Driven Architecture | API & Integration Architecture | Platform Engineering

AWS Cloud & DevOps - Kubernetes | Docker | Temporal | GitHub Actions | EC2 | ECS/EKS | Lambda | API Gateway | AWS Services

AI & GenAI - OpenAI | Gemini | Claude | LangGraph | LangChain | Langfuse | MCP | Agentic AI | Prompt Engineering | RAG | Knowledge Graphs | SBERT

Data & Messaging - Apache NiFi | RabbitMQ | Kafka

Databases - GraphDB | Neo4j | PostgreSQL | Aurora PostgreSQL | MongoDB

Frameworks - Spring Boot | Spring Cloud | Hibernate | FastAPI

PREVIOUS EXPERIENCE

Oracle | Applications Engineer | 2019–2022

- Modernized Oracle Fusion Procurement applications by migrating Oracle ADF/JSF to Oracle JET.
- Designed REST APIs supporting Oracle Fusion ERP workflows.

ComakeIT | Software Engineer | 2017–2019

- Designed RESTful microservices using Java and Spring Boot.
- Built automated testing frameworks using Selenium, Cucumber, and JUnit.

HCL Technologies | Software Engineer | 2015–2017

- Developed enterprise applications using Java, Spring, REST, SOAP, JPA, and JDBC.
- Improved data processing performance through multithreaded programming.